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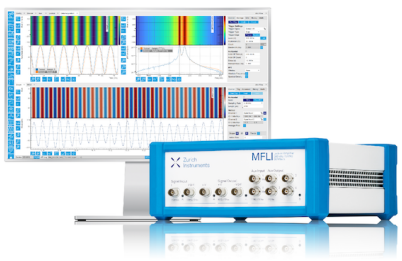
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Accurate *ab initio* determination of the adiabatic potential energy function and the Born–Oppenheimer breakdown corrections for the electronic ground state of LiH isotopologues

Filip Holka,[1,2] Péter G. Szalay,[3,a)] Julien Fremont,[1] Michael Rey,[1]
Kirk A. Peterson,[4] and Vladimir G. Tyuterev[1,b)]
[1]*Groupe de Spectrométrie Moléculaire et Atmosphérique, Reims University, F-51687 Reims, France*
[2]*Faculty of Materials Science and Technology, Slovak University of Technology, Paulínska 16, 917 24 Trnava, Slovak Republic*
[3]*Institute of Chemistry, Eötvös Loránd University, H-1117 Budapest, Hungary*
[4]*Department of Chemistry, Washington State University, Pullman, Washington 99164-4630, USA*

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High level *ab initio* potential energy functions have been constructed for LiH in order to predict vibrational levels up to dissociation. After careful tests of the parameters of the calculation, the final adiabatic potential energy function has been composed from: (a) an *ab initio* nonrelativistic potential obtained at the multireference configuration interaction with singles and doubles level including a size-extensivity correction and quintuple–sextuple ζ extrapolations of the basis, (b) a mass–velocity-Darwin relativistic correction, and (c) a diagonal Born–Oppenheimer (BO) correction. Finally, nonadiabatic effects have also been considered by including a nonadiabatic correction to the kinetic energy operator of the nuclei. This correction is calculated from nonadiabatic matrix elements between the ground and excited electronic states. The calculated vibrational levels have been compared with those obtained from the experimental data [J. A. Coxon and C. S. Dickinson, J. Chem. Phys. **134**, 9378 (2004)]. It was found that the calculated BO potential results in vibrational levels which have root mean square (rms) deviations of about 6–7 cm^{-1} for LiH and $\sim 3\,cm^{-1}$ for LiD. With all the above mentioned corrections accounted for, the rms deviation falls down to $\sim 1\,cm^{-1}$. These results represent a drastic improvement over previous theoretical predictions of vibrational levels for all isotopologues of LiH. © *2011 American Institute of Physics*. [doi:10.1063/1.3555758]

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I. INTRODUCTION

A calculation of molecular potential energy surfaces up to the dissociation limit and accurate theoretical predictions of highly excited vibrational states is relevant for various applications of physical chemistry and spectroscopy, including high-temperature emission measurements, laser fluorescence, and multiple-resonance laser excitation experiments, but are rather challenging for theory. Despite considerable recent progress, the accuracy of rovibrational *ab initio* calculations near the dissociation limit is still quite far from spectroscopic requirements, except for one- and two-electron systems.[1–8]

The lightest heteronuclear stable compound, LiH with only four electrons, has been often investigated by both quantum chemistry methods[9–17] and spectroscopy[18–23] during past decades. In the former case, theoretical calculations have always reflected the state of the art in *ab initio* electronic structure calculations. With regards to the present study, i.e., the potential energy curve (PEC) of LiH, the first two papers to note here are from 1981 by Partridge and Langhoff[9] and by Jönsson *et al.*[10] The former calculations consisted of a multireference configuration interaction (MR-CI) wave function and about 60 Slater-type basis functions with errors of the calculated vibrational energies of about 5–10 cm^{-1} for the first 11 levels. The latter work included a two-determinant reference configuration interaction (CI) with singles and doubles (CISD) wave function with a 7s5p3d/6s3p basis that resulted in considerably larger errors. Shortly later, Bishop and Cheung[11] obtained the adiabatic correction from theory for the first time. They used a selected CI wave function with a small basis set to obtain this quantity. In 1988 Jensen and Yarkony[12] re-evaluated the adiabatic correction with a larger MR-CI wave function. It was in 1992 when Gadéa and co-workers[13] started a series of calculations on LiH. Here valence full-CI calculations with a core polarization potential were performed with a fairly large 7s6p3d1f/5s3p2d basis set. Later their calculations were augmented by an adiabatic correction[14] and also the nonadiabatic (NAD) corrections were considered.[15] In 2006, Gadéa and Leininger[16] repeated the calculations using an even larger basis (27s17p14d6f2g/20s11p7d4f2g). Of particular interest are the calculations by Li and Paldus[17] who used multireference coupled cluster (MR-CC) methods and cc-pVXZ basis sets with $X = \mathrm{T,Q,5}$ along with a three point extrapolation formula to the approximate basis set limit. The applied MR-CC method is very accurate, delivering nearly full-CI results for LiH. By incorporating the experimentally obtained adiabatic correction function for the H atom from Chan *et al.*,[18] an adiabatic correction was also included in that study. Although the PEC by Li and Paldus[17] can be considered to be the most accurate one available before the present study, there are

a)Electronic mail: szalay@chem.elte.hu.
b)Electronic mail: vladimir.tyuterev@univ-reims.fr.

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clear shortcomings, e.g., the use of the family of correlation consistent basis sets without core correlation functions, but correlating also the core electrons in the wave function.

A very large sample of rovibrational transitions involving X and A electronic states have been measured for four isotopologues, ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD, leading to the determinations of vibrational levels very close to the dissociation limit. A detailed description of the spectroscopic work, including a wealth of the experimental data and empirically fitted models, has been given by Coxon and Dickinson.[19] One of the motivations for these extended measurements and calculations was an empirical evaluation of isotopically dependent Born–Oppenheimer (BO) breakdown effects expected for these light systems. Note that the error of the best available *ab initio* prediction[17] for high energy vibrations in LiH amounts to $\approx$30 cm^{-1}, whereas empirical estimations for non-BO contributions[19] represent only a fraction of this error. An investigation of the importance of various theoretical contributions and an improvement of the accuracy of vibrational predictions for light molecules thus remains an open question.

A separation of electronic and nuclear variables accounting for adiabatic and nonadiabatic terms has been considered in many studies.[5,24–28] In this work, the Bunker–Moss[25] formulation was used for the Hamiltonian of nuclear vibrations of diatomic molecules in the electronic ground state:

$$\begin{aligned} H^{\mathrm{VIB}} &= \frac{1}{2\mu} P_R(1+\beta(R))P_R + U^{\mathrm{BO}}(R) + \\ &\quad + U^{\mathrm{DBOC}}(R) + U^{(2)}(R) + \ldots, \\ P_R &= -i\hbar\frac{\partial}{\partial R}. \end{aligned} \tag{1}$$

Here R is the internuclear distance, P_R is the conjugate vibrational momentum, and μ is the reduced nuclear mass. $U^{\mathrm{BO}}(R)$ represents the BO (clamped nuclei) potential energy function independent of nuclear mass and $U^{\mathrm{DBOC}}(R)$ is the mass-dependent diagonal BO breakdown correction (DBOC) to this function. According to Handy *et al.*[29,30] and Kutzelnigg,[31] the DBOC correction for the electronic ground state ψ_0 can be defined in terms of diagonal matrix elements of the second derivatives with respect to nuclear Cartesian coordinates:

$$U^{\mathrm{DBOC}}(R) = -\frac{\hbar}{2}\sum_A \sum_{X_A} \frac{1}{M_A}\langle\psi_0|\frac{\partial^2}{\partial X_A^2}|\psi_0\rangle \tag{2}$$

with M_A being the nuclear mass of A. The sum runs over all Cartesian coordinates X_A of the nuclei A in the considered molecule and the integration goes over electronic coordinates only. The function $\beta(R)$ in Eq. (1) represents a nonadiabatic correction to the kinetic energy operator of nuclear vibrations involving the coupling matrix elements of the first nuclear derivative between the ground ψ_0 and excited ψ_n electronic states:

$$\beta(R) = -\frac{2\hbar^2}{\mu}\sum_n \frac{\langle\psi_0|\frac{\partial}{\partial R}|\psi_n\rangle^2}{U_0^{\mathrm{BO}}(R) - U_n^{\mathrm{BO}}(R)}. \tag{3}$$

The term $U^{(2)}(R)$ is a second order NAD correction to the potential energy function which will not be considered in the present study. The procedure for the *ab initio* calculation of DBOC corrections has been reviewed in Refs. 32 and 33, while that for the *ab initio* calculation of nonadiabatic corrections to the effective rovibrational Hamiltonian has been recently considered in detail by Schwenke with applications to the H_2^+, HD^+, H_2, and H_2O molecules.[5,34]

The goal of the present study is to obtain an improved adiabatic potential energy function and the Born–Oppenheimer breakdown corrections for the electronic ground state of LiH from state of the art *ab initio* calculations and to investigate various factors influencing the accurate calculation of theoretical vibrational levels at high energies approaching the dissociation limit. The paper is structured as follows. Section II is devoted to electronic structure calculations with comparisons to previous *ab initio* studies of the BO potential function and DBOC corrections. In these calculations we use a higher level of theoretical treatment than earlier studies in all respects. As for the electron correlation we use a nearly full-CI quality description (large MR-CISD with Davidson-type correction) considering all electrons. The basis set used is up to sextuple-ζ quality including core functions and complete basis set (CBS) extrapolation. Finally, we show that the inclusion of the DBOC results in a substantial improvement of the potential energy function. In Sec. III, we consider nonadiabatic couplings and derive the $\beta(R)$ of Eq. (3). Section IV describes vibrational calculations involving BO, relativistic, DBOC, and $\beta(R)$ corrections, where we also compare calculations using the nuclear and atomic masses. Conclusions and the discussion of results are given in Sec. V.

II. ELECTRONIC STRUCTURE CALCULATIONS

The goal of the first part of the present work is to construct the adiabatic *ab initio* potential energy function as accurately as possible. The accuracy is controlled by investigating the results of a well defined series of calculations, in this respect following procedures used for the accurate calculation of heats of formation, such as the high accuracy extrapolated *ab initio* thermochemistry (HEAT),[35] the Weizmann x techniques,[36] or the focal point analysis.[37] The accuracy will be monitored by comparing the vibrational energies associated with potential energy curves of the given theoretical levels. This procedure is preferred over the inspection of the electronic energy at different geometries since it directly provides a measure of accuracy for the quantities we wish to describe. During the course of this study it was observed that vibrational energies are very sensitive to the details of the surface, and simple inspection of the electronic energy is not sufficient to control the accuracy. Thus, for all tested levels of theory the energy (or other quantities, see later) were calculated along a grid of 243 nuclear configurations and the corresponding numerically exact vibrational energy levels [with respect to zero point energy (ZPE)] were obtained. The method used for these vibrational calculations will be described later in Sec. IV. By construction, the adiabatic potential energy curve has three components: (a) a nonrelativistic BO surface; (b) a DBOC; and (c) the contribution of relativistic effects. These are separately discussed below.

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A. Nonrelativistic Born–Oppenheimer function

Due to the small number of electrons it was possible to reach a high level for both electron correlation and basis set. All calculations on the BO surface have been performed by the COLUMBUS suite of programs[38] except the full-CI calculations, which have been performed with the MRCC program of Kállay[39] coupled to the CFOUR suite.[40]

The MR-CISD wave function was built on the top of a complete active space (CAS) reference space including 8σ, 3π, 1δ orbitals (8-3-1) and all four electrons [16 orbitals and 4 electrons resulting in 1185 configuration state functions (CSFs) in C_{2v} symmetry]. All single and double excitations out of these functions were included in the MR-CI wave function resulting in up to about 3.5 million configurations with the largest basis sets. The orbitals were obtained in a multiconfigurational self-consistent field (MCSCF) procedure using the same reference space. The CI energy was augmented by Davidson-type corrections.[41] Two types were used: the original one suggested by Davidson for the single reference case[42,43] (referred to hereafter as $+Q_D$) and the somewhat more sophisticated Pople-variant[44] (referred to as $+Q_P$). As will be obvious below, the size-consistency error is very small for this MR-CISD wave function, therefore, we do not find it necessary to apply more sophisticated coupled electron pair approximation-type methods, such as multireference averaged quadratic coupled cluster[45] or multireference averaged coupled pair functional.[46]

We start by testing the accuracy of the BO potential with the comparison of the vibrational energy levels calculated using different electron correlation methods and the cc-pwCVQZ (Refs. 47 and 48) basis set. The first observation is that all three methods (MR-CISD, MR-CISD+Q_D, and MR-CISD+Q_P) give very similar results (see Table I and Fig. 1), the largest deviation among them amounts to only a fraction of a wavenumber. The $+Q_D$ correction seems to overestimate the missing higher excitation effects: compared to MR-CISD, the sign of the error with respect to full-CI is flipped and its magnitude is only slightly smaller. The $+Q_P$ correction, on the other hand, agrees perfectly with the full-CI results, the largest error being about 0.1 cm^{-1}. The better performance of the $+Q_P$ corrections has been already predicted by theoretical analysis (see, e.g., Ref. 41) and linked to the consideration of exclusion principle violating terms in the later method. One is tempted to say that the size-consistency correction does not seem to be important here, but (as seen later) very small contributions of other types will be included, therefore, it seems to be appropriate to choose the most accurate MR-CISD+Q_P variant for all later calculations. Since the good agreement of the MR-CISD, MR-CISD+Q_D, and MR-CISD+Q_P results applies also for larger basis sets where full-CI calculations are not feasible, it can be safely concluded that the remaining uncertainty due to missing electron correlation is not expected to be larger than 0.1–0.2 cm^{-1} for all vibrational levels.

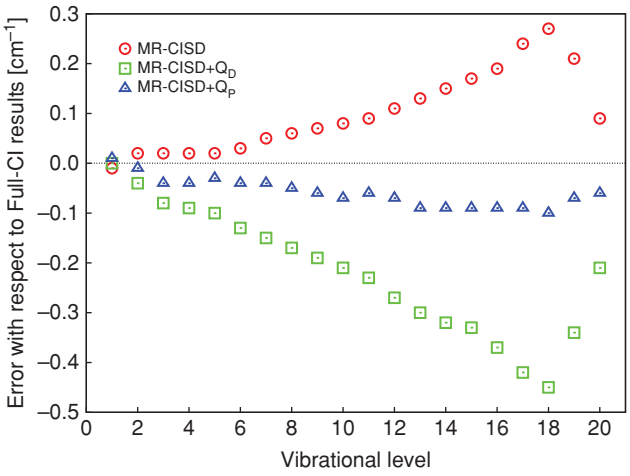


FIG. 1. Convergence of the vibrational levels of ^{7}LiH calculated using different electron correlation methods with respect to the full-CI results. See Sec. II for details of the *ab initio* calculations.

TABLE I. Vibrational energies (with respect to ZPE), dissociation energy (cm^{-1}), and equilibrium bond length (bohr) of ^{7}LiH calculated from potential energy curves obtained by different correlated methods and the cc-pwCVQZ basis set. See Sec. II for details of the *ab initio* calculations.

v	MR-CISD	MR-CISD+Q_D	MR-CISD+Q_P	Full-CI
1	1362.13	1362.12	1362.11	1362.12
2	2679.73	2679.79	2679.76	2679.75
3	3953.82	3953.92	3953.88	3953.84
4	5185.24	5185.35	5185.30	5185.26
5	6374.71	6374.83	6374.76	6374.73
6	7522.84	7523.00	7522.91	7522.87
7	8630.12	8630.32	8630.21	8630.17
8	9696.88	9697.11	9696.99	9696.94
9	10 723.22	10 723.48	10 723.35	10 723.29
10	11 709.01	11 709.30	11 709.16	11 709.09
11	12 653.79	12 654.11	12 653.94	12 653.88
12	13 556.68	13 557.06	13 556.86	13 556.79
13	14 416.32	14 416.75	14 416.54	14 416.45
14	15 230.67	15 231.14	15 230.91	15 230.82
15	15 996.80	15 997.30	15 997.06	15 996.97
16	16 710.62	16 711.18	16 710.90	16 710.81
17	17 366.53	17 367.19	17 366.86	17 366.77
18	17 957.01	17 957.73	17 957.38	17 957.28
19	18 472.13	18 472.68	18 472.41	18 472.34
20	18 898.84	18 899.14	18 898.99	18 898.93
D_e	20 175.50	20 175.40	20 174.90	20 174.90
R_e	3.01383	3.01378	3.01377	3.01381

Having set the proper electronic structure method, the next step is to find the one-electron basis that can give similar accuracy. Since correlation of the core electrons of Li is explicitly taken into account by the present MR-CISD wave function, the newly developed cc-pwCVXZ basis sets of Peterson and co-workers[47,48] with cardinal numbers $X = $ Q,5,6 were chosen for Li, while for H the corresponding correlation consistent cc-pVXZ basis sets[49] were applied. In what follows, the basis will be referred to as cc-pwCVXZ keeping in mind that the cc-pVXZ basis is used for hydrogen. Note that the largest calculations with the cc-pwCV6Z basis involve 349 basis functions. Basis set extrapolation of the correlation energy to the CBS limit can be best performed by the two-point formula:[50,51]

$$E_{\text{corr}}^{X} = E_{\text{corr}}^{\text{CBS}} + \frac{\text{const}}{X^3}, \tag{4}$$

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TABLE II. Vibrational energies (with respect to ZPE), dissociation energy (cm^{-1}), and equilibrium bond length (bohr) of ^{7}LiH calculated for potential energy curves obtained with the MR-CISD+Q_P method and different cc-pwCVXZ (X = Q,5,6) basis sets [the abbreviation XZ (X = Q,5,6) is used for simplicity] with/without extrapolation to CBS included [the abbreviation (XY) (X,Y = Q,5,6) is used where X and Y denote cardinal numbers of basis sets used in the extrapolation]. See Sec. II for details of the *ab initio* calculations and extrapolations.

	Without extrapolation				With extrapolation	
v	5Z[a] 5Z[b]	6Z[a] QZ[b]	6Z[a] 5Z[b]	6Z[a] 6Z[b]	6Z[a] (Q5)[b]	6Z[a] (56)
1	1360.6	1361.6	1360.5	1360.7	1359.3	1361.0
2	2676.6	2678.8	2676.4	2676.6	2673.8	2677.0
3	3948.9	3952.3	3948.5	3948.6	3944.4	3948.8
4	5178.2	5183.1	5177.7	5177.5	5172.0	5177.4
5	6365.4	6371.8	6364.7	6364.2	6357.3	6363.4
6	7511.1	7519.1	7510.2	7509.1	7500.9	7507.6
7	8615.8	8625.6	8614.7	8613.0	8603.3	8610.6
8	9679.8	9691.5	9678.5	9676.0	9664.9	9672.6
9	10 703.4	10 716.9	10 701.8	10 698.5	10 685.9	10 693.9
10	11 686.3	11 701.9	11 684.5	11 680.4	11 666.3	11 674.6
11	12 628.3	12 645.9	12 626.3	12 621.3	12 605.8	12 614.3
12	13 528.7	13 548.2	13 526.5	13 520.6	13 503.7	13 512.5
13	14 386.1	14 407.3	14 383.8	14 377.2	14 359.1	14 368.0
14	15 198.9	15 221.4	15 196.5	15 189.2	15 170.3	15 179.3
15	15 964.4	15 987.5	15 961.9	15 954.3	15 935.0	15 943.8
16	16 678.9	16 701.6	16 676.4	16 668.8	16 650.0	16 658.3
17	17 337.3	17 358.2	17 334.9	17 327.8	17 310.5	17 317.9
18	17 932.9	17 949.8	17 930.7	17 924.7	17 910.5	17 916.4
19	18 456.3	18 466.5	18 454.4	18 450.6	18 441.3	18 445.3
20	18 895.6	18 895.3	18 894.1	18 893.8	18 891.7	18 893.2
D_e	20 241.2	20 182.1	20 242.1	20 268.8	20 305.1	20 305.4
R_e	3.01402	3.01393	3.01410	3.01402	3.01428	3.01390

[a]Basis set used for the calculation of noncorrelated energy (2×2 CAS).
[b]Basis set used for the calculation of correlation energy.

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where E^X_{corr} is the correlation energy calculated with the cc-pwCVXZ basis (X denoting its cardinal number), E^{CBS}_{corr} is the (extrapolated) CBS energy. However, the correlation energy is not well defined for a multireference wave function, in particular, if a large range of bond distances are considered. Therefore, we define the "uncorrelated" situation by the 2 × 2 CAS wave function that is able to describe the bond dissociation properly. We will refer to this part of the total electronic energy as the "noncorrelated energy." The correlation energy is then defined as the total energy calculated by the MR-CISD+Q_P method minus the noncorrelated energy:

$$E^X_{corr} = E^X_{MR-CISD+Q_P} - E^X_{2x2CAS} \quad (5)$$

It is well known[52] from single-reference calculations that the Hartree–Fock energy converges faster with the basis set (exponential decrease with cardinal number) than the correlation energy. Since the cc-pwCV6Z basis seems to be large enough to provide converged results for the "uncorrelated" case, the noncorrelated energy is not extrapolated and the cc-pwCV6Z energy is used throughout. This assumption seems to work well and the remaining error due to this contribution is not expected to be larger than 0.1–0.3 cm^{-1}. This can be clearly seen by comparing the second and the fourth column in Table II, where the underlying potentials differ only in that the reference part of the energy was calculated with cc-pwCV5Z and cc-pwCV6Z basis sets, respectively, while for the correlation energy in both cases the cc-pwCV5Z set was used.

To shorten the abbreviation, the following notation is introduced for the basis sets and extrapolated results: XZ refers to the basis set cc-pwCVXZ with X = Q,5,6; XZ/YZ means that the noncorrelated energy has been calculated with the XZ basis with the correlation energy by the YZ basis. Instead of XZ, (Q5) and (56) are used if the corresponding energy comes from quadruple–quintuple or quintuple–sextuple extrapolation.

In Fig. 2, the potential energy curves calculated with different basis sets, as well as extrapolations, are compared. In this figure the relative differences are plotted with respect to the 6Z/(56) PEC (Ref. 53) scaled to a common dissociation limit. All calculations have been performed at the MR-CISD+Q_P level. We see a smooth convergence, although the change from 6Z/6Z to 6Z/(56) (reference line) is still substantial. On the other hand, the (Q5) extrapolation—although reducing the error—seems to overestimate the basis set limit, giving lower energies than the (56) extrapolation. The shape of the curves show that the basis set error depends substantially on the nuclear distance, the maximum deviation from the reference line is in all cases situated at about 6 bohr. Therefore, we expect that the vibrational energies will vary substantially with the basis set size as well as from the basis

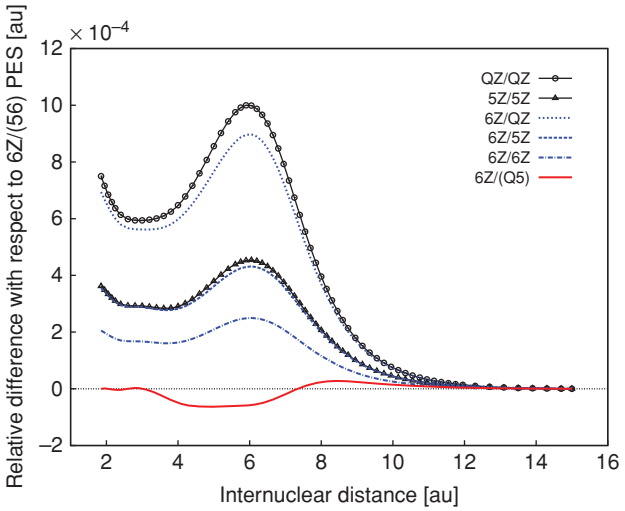


FIG. 2. Relative differences of ^{7}LiH potential energy curves (hartree), scaled to the dissociation energy limit, calculated with the MR-CISD+Q_P method and different cc-pwCVXZ (X = Q,5,6) basis sets with respect to the highest level 6Z/(56) PEC. See Sec. II for details of the *ab initio* calculations.

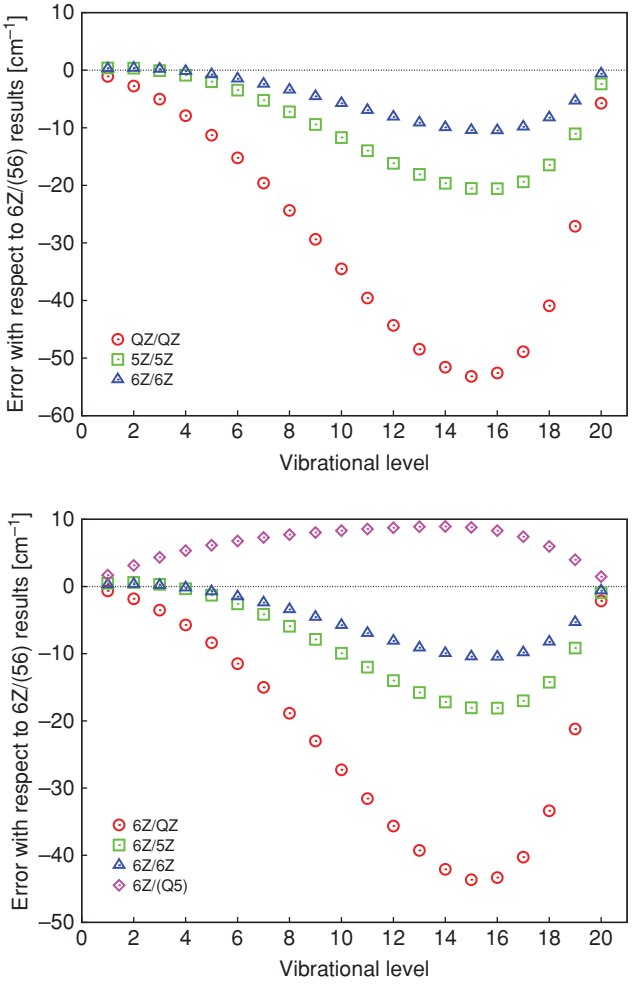


FIG. 3. Convergence of the vibrational levels of ^{7}LiH calculated for potential energy curves obtained with the MR-CISD+Q_P method and different cc-pwCVXZ (X = Q,5,6) basis sets with respect to the highest level 6Z/(56) results. Upper panel: calculations with conventional basis sets. Lower panel: "noncorrelated" part of the energy obtained with the cc-pwCV6Z basis. See Sec. II for details of the *ab initio* calculations.

set extrapolation. The maximal sensitivity (largest changes) are expected for those vibrational levels having turning points around 6 bohr (see later).

The comparison of vibrational levels obtained with the different basis sets and the MR-CISD+Q_P method are collected in Table II and Fig. 3. First inspection of these shows that the vibrational levels in most cases decrease with the enlargement of the basis while the D_e value increases. This effect is shown in the upper panel of Fig. 3 in closer detail. One should notice that the errors of vibrational energies obtained with the cc-pwCVQZ basis set [with respect to the best BO PEC (Ref. 53)] is in an acceptable range for the first level only. Going to higher vibrational levels, the error increases, reaching a maximum of ≈55 cm^{-1} around $\mathbf{v} = 15$. The larger basis sets (with $X = 5,6$) provide acceptable accuracy up to $\mathbf{v} = 5$–6, however, the error for higher vibrational states is still significant (≈10 cm^{-1} for $\mathbf{v} = 15$ in cc-pwCV6Z basis).

Effect of the basis set extrapolation of the correlation energy on the calculated vibrational energies is shown in the lower panel of Fig. 3 (see also Table II). The (Q5) extrapolation (i.e., using X = Q and 5 in the extrapolation) has a large effect, decreasing the energies by 1 cm^{-1} for $\mathbf{v} = 1$ to almost 20 cm^{-1} for $\mathbf{v} = 15$ (see lower panel of Fig. 3 and the fifth and sixth column in Table II). Comparing the 6Z/(Q5) extrapolated values to the cc-pwCV6Z results (deviation of 1.4 and 19 cm^{-1} for $\mathbf{v} = 1$ and $\mathbf{v} = 15$, respectively), one observes that the extrapolated values are apparently largely overestimated, since the cc-pwCV5Z and cc-pwCV6Z results differ by much less (0.2 and 8 cm^{-1} for the first and 15th level, respectively). The 6Z/(Q5) values also overestimate the final BO 6Z/(56) vibrational energies (up to ≈10 cm^{-1} for $\mathbf{v} = 15$). Moreover, for low vibrational states, the errors are even bigger [with respect to the 6Z/(56) results] than obtained with the pure cc-pwCV6Z basis set. This indicates that the cc-pwCVQZ basis set is far from being converged, and thus the extrapolation using such a basis may introduce large inaccuracies. Considering the (56) extrapolation, one observes a much milder effect, the maximum change is about 10 cm^{-1} with respect to the cc-pwCV6Z basis.

Therefore, in the final calculations the 6Z/(56) basis will be used, i.e., cc-pwCV6Z for the noncorrelated energy and quintuple–sextuple extrapolation for the correlation energy. The error of the extrapolated BO value should not be larger than 0–20% of the correction, thus more than a 2 cm^{-1} error is not expected for even the higher vibrational levels, while the inaccuracy of the lowest 4-5 levels is probably smaller than 0.2 cm^{-1}. This assumption is also supported by the very good agreement of the final calculated vibrational levels to the experimental ones (see later).

Interestingly, the extrapolation seems to work better for the dissociation energy (see Table II), (Q5) and (56) values practically match although they differ substantially from the 6Z value by almost 40 cm^{-1}. It is concluded that the extrapolation is absolutely necessary for the dissociation energy even in the case of sextuple-ζ basis set. The equilibrium bond length is well converged to 3.0139 ± 0.0001 bohr.

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B. Adiabatic DBOC correction

The diagonal Born–Oppenheimer correction has been calculated analytically using the procedure developed in Ref. 32. This procedure requires the calculation of a derivative wave function parameter, so the computational task is equivalent to the calculation of harmonic vibrational frequencies by analytical techniques.[54] Since the DBOC values are needed along a large interval of the bond distance, multireference calculations are necessary. Encouraged by the accuracy of the applied MR-CISD wave function, it was decided to use the same MR-CISD wave function (8-3-1 CAS as reference) to obtain the DBOC. To that end the MRCC program by Kállay[39] coupled to the CFOUR suite[40] was used. Due to program limitations the orbitals were obtained from Hartree–Fock calculations: since the full-CI energy is invariant with respect to the choice of orbitals and the wave function used here is nearly of full-CI quality, a sizable error from this restriction is not expected. As has been shown in Ref. 32, core correlation effects are important for the DBOC, therefore, the cc-pwCVTZ basis (cc-pVTZ on hydrogen) was primarily used for these calculations. Since the DBOC calculations are quite demanding, the effect of reducing the reference space to 5σ, 2π (5-2), and even 4σ, 1π (4-1) was investigated. The smaller reference space also enabled a test of the basis set convergence by performing cc-pwCVQZ calculations at the MR-CI level.

The results are summarized in Table III where the change of the vibrational energy when the DBOC[53] is added to the nonrelativistic BO potential [6Z/(56) extrapolated MR-CISD+Q_P level] is listed. The values calculated at the different levels of theory agree quite nicely; the results of the 8-3-1 reference space differ only by 0.2 cm^{-1} for the highest levels and the dissociation energy when compared with the smaller reference spaces. The effect of the basis set can be obtained by comparing the (4-1) calculations with cc-pwCVTZ and cc-pwCVQZ basis sets: the difference is about 0.1 cm^{-1} at maximum. Therefore, the 8-3-1 CAS reference with the cc-pwCVTZ basis set will be used in subsequent calculations.

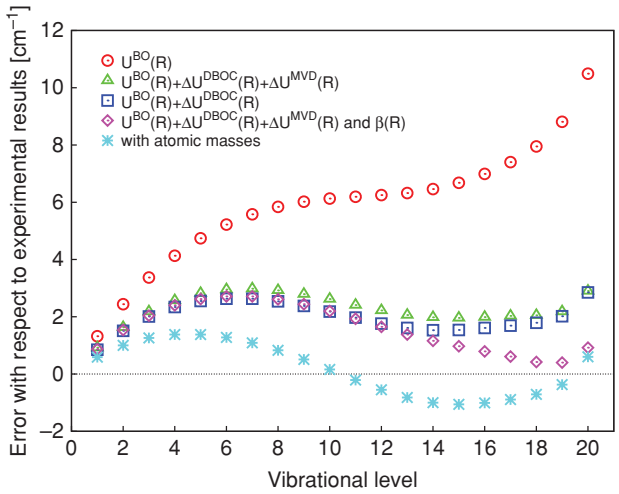


FIG. 4. Comparison of the calculated vibrational levels of ^{7}LiH with the experimental determinations (Ref. 19). The figure also shows the importance of the different contributions [DBOC, relativistic effect, nonadiabatic contribution (β)] and the use of atomic versus nuclear masses. See text for more explanations.

TABLE III. Contribution of the DBOC to the vibrational levels and dissociation energy (cm^{-1}) of ^{7}LiH obtained with the MR-CISD+Q_P method using different basis sets and different reference CAS spaces. See Sec. II for details of the *ab initio* calculations.

	Reference CAS space				
v	$4\sigma,1\pi$[a]	$5\sigma,2\pi$[a]	$8\sigma,3\pi,1\delta$[a]	$4\sigma,1\pi$[b]	emp.[c]
1	−0.47	−0.47	−0.47	−0.46	−0.44
2	−0.92	−0.92	−0.92	−0.91	−0.92
3	−1.36	−1.36	−1.36	−1.35	−1.39
4	−1.79	−1.78	−1.78	−1.77	−1.81
5	−2.20	−2.19	−2.19	−2.18	−2.20
6	−2.60	−2.59	−2.58	−2.58	−2.58
7	−2.98	−2.97	−2.95	−2.96	−2.96
8	−3.35	−3.34	−3.31	−3.33	−3.33
9	−3.70	−3.69	−3.64	−3.69	−3.68
10	−4.03	−4.02	−3.95	−4.03	−4.00
11	−4.33	−4.32	−4.23	−4.35	−4.30
12	−4.61	−4.61	−4.48	−4.65	−4.60
13	−4.87	−4.86	−4.71	−4.93	−4.89
14	−5.10	−5.10	−4.93	−5.19	−5.00
15	−5.33	−5.34	−5.14	−5.44	−4.87
16	−5.59	−5.61	−5.39	−5.70	−5.02
17	−5.93	−5.96	−5.71	−6.01	−5.38
18	−6.41	−6.44	−6.16	−6.46	−5.35
19	−7.08	−7.10	−6.79	−7.12	−6.94
20	−7.97	−8.00	−7.64	−8.04	−7.50
D_e	−12.29	−12.25	−12.03	−12.56	−12.00

[a]Calculated with cc-pwCVTZ basis set.
[b]Calculated with cc-pwCVQZ basis set.
[c]Empirical adiabatic DBOC correction by Coxon and Dickinson (Ref. 19).

Figure 4 includes the overall effect of different contributions to the vibrational energies including DBOC. It is seen from this figure that the effect of the DBOC is smaller than 1 cm^{-1} for the lowest states and increases monotonically with the increasing vibrational quantum number. For the 20th state it reaches 8 cm^{-1}, thus clearly not negligible. Its effect on the dissociation energy (see Table III) is also substantial, amounting to about 12 cm^{-1}.

In Table III, the contributions of the empirically deduced DBOC function by Coxon and Dickinson[19] on the vibrational energies of ^{7}LiH are also given. In Fig. 5, this empirical DBOC function is compared with the final *ab initio* one. Qualitatively, the agreement is very good, although a quantitative comparison of *ab initio* radial functions with empirically fitted ones should be taken with caution. As shown by Watson,[26,27] there is a fundamentally unavoidable problem of ambiguities in empirical fits of radial BO, DBOC, and nonadiabatic functions to experimental isotopic transitions. Only certain combinations of these radial functions could be empirically determined. In this sense the comparison of final vibrational levels including DBOC contributions is more instructive.

As will be seen later in the discussion, the calculated vibrational levels are in very good agreement with the experiment for all isotopologues (maximum error is only 1–2 cm^{-1}). This implies that the calculated DBOC accounts nearly perfectly for the mass-dependent contributions. Therefore, it is instructive that the calculated total DBOC (sum of

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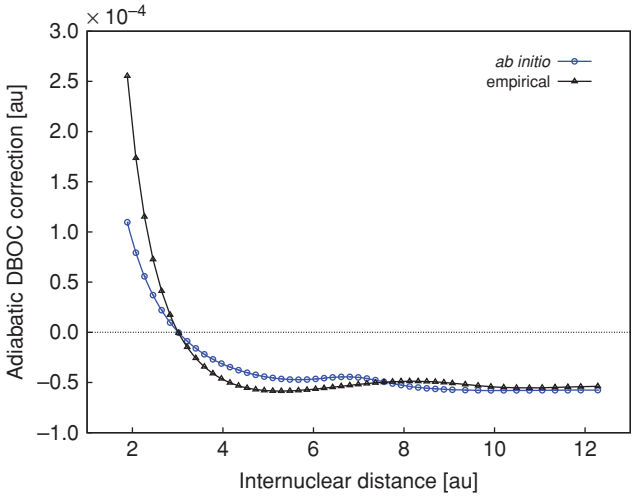


FIG. 5. Comparison of the *ab initio* and empirical DBOC functions. The *ab initio* function has been obtained with the MR-CISD method using a 8-3-1 reference space and the cc-pwCVTZ basis. The empirical function is plotted using the parameters of Ref. 19. Curves are shifted to zero energy value near the equilibrium bond length (3 bohr).

atomic contributions) curve runs similarly to the empirical one,[19] but there is a large discrepancy for the atomic DBOC contributions (Fig. 6). This deficiency of empirical models has been already mentioned in previous studies (see Ref. 19 for discussion) and supports the statement of Watson[22,26,27] concerning the ambiguity of empirical DBOC determinations. This is particularly true because in most empirical fits the nuclear Hamiltonian is constrained in such a way that the other mass-dependent term, the β (R) NAD contribution, is removed.

It can be concluded in passing that the inclusion of the DBOC is absolutely necessary to obtain the required wavenumber accuracy. The remaining uncertainty in the calculated DBOC correction obtained at the present standard level (8-3-1 reference space and cc-pwCVTZ basis) is expected to be about 0.01 cm^{-1} for the lowest levels and not larger than 0.2 cm^{-1} for the highest vibrational states.

C. Relativistic correction

Seeking the most accurate potential energy curve one should also investigate the contribution of relativistic effects, even if these are not expected to be large for light systems like LiH. The well established and simple way to account for the relativistic corrections is based on the Dirac–Coulomb–Breit–Pauli approximation. Within the framework of this approach the relativistic effects are treated as perturbations and determined as the first-order corrections to the nonrelativistic energy. The correction is calculated by adding expectation values of "mass–velocity" and Darwin operators with the respective nonrelativistic one-electron density (which is usually referred to as the MVD correction). In order to realize the calculation of MVD relativistic contributions to the potential energy curve of LiH, the internally contracted MR-CISD (icMR-CISD) method of Werner and Knowles[55,56] was used with the same 4 electrons, 15 orbitals CAS reference space as in the nonrelativistic (uncontracted) MR-CI calculations discussed above. Since the vibrational levels are only negligibly affected by relativistic effects (see later), the use of the internally contracted scheme is not expected to influence the results. Also the atomic basis sets were the same in these calculations, i.e., cc-pwCVXZ (X=T,Q,5). The MOLPRO software package (version 2006.1) (Ref. 57) was used in these calculations.

The MVD relativistic effects on the potential energy curve obtained with the different basis sets are shown in Fig. 7. The overall magnitude of the relativistic contribution is the same order ($\approx 10^{-4}$ hartree) as, e.g., the DBOC correction or the effect of basis set used in calculation of the PEC (compare to Figs. 5 and 2). Nevertheless, the influence on the vibrational levels is only up to about 0.5 cm^{-1} for $\mathbf{v} = 11$–14 (see Table IV and Fig. 4) compared to the $\approx$5 cm^{-1} in the case

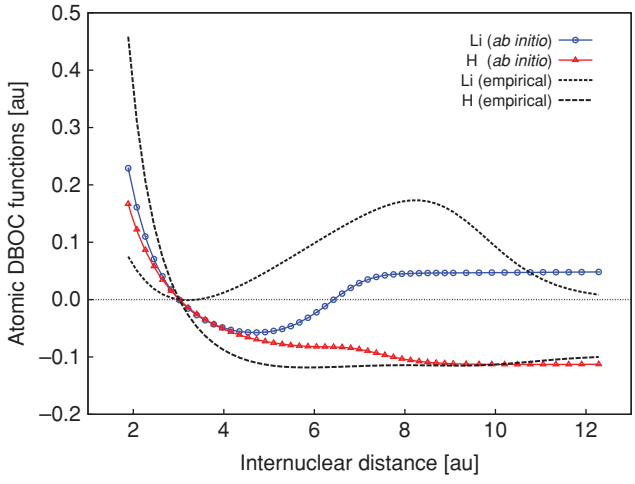


FIG. 6. Comparison of the *ab initio* and empirical atomic DBOC functions. The *ab initio* function has been obtained with the MR-CISD method using a 8-3-1 reference space and the cc-pwCVTZ basis. The empirical functions are plotted using the parameters of Ref. 19. Curves are shifted to zero energy value near the equilibrium bond length (3 bohr).

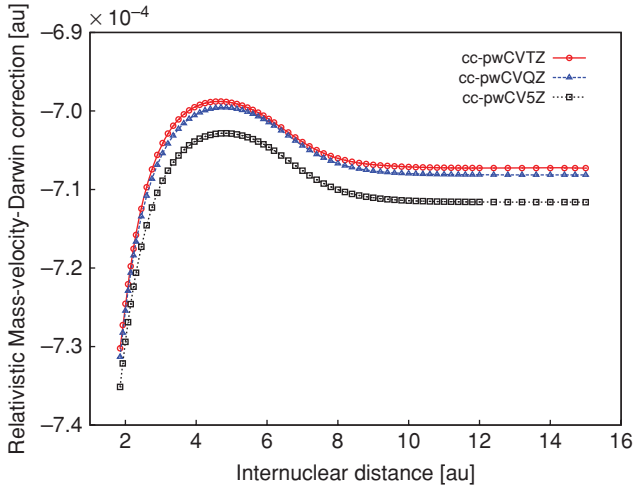


FIG. 7. Perturbative mass–velocity-Darwin relativistic correction to the ground electronic state potential energy curve of LiH calculated with the internally contracted MR-CISD method using different cc-pwCVXZ (X = T,Q,5) basis sets.

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TABLE IV. Contributions (in cm^{-1}) to the vibrational energy levels using nuclear masses. All data refer to the most abundant isotopologue (^{7}LiH).

v	Nonrelativistic BO[a]	Δ_{DBOC}[b]	Δ_{MVD}[c]	Adiabatic[d]	Empirical[e]	Calc.-Emp.
1	1361.03	− 0.46	0.06	1360.63	1359.71	0.92
2	2677.00	− 0.91	0.12	2676.21	2674.56	1.65
3	3948.84	− 1.35	0.18	3947.67	3945.47	2.20
4	5177.40	− 1.77	0.23	5175.86	5173.28	2.58
5	6363.47	− 2.18	0.27	6361.56	6358.73	2.83
6	7507.69	− 2.58	0.31	7505.42	7502.47	2.95
7	8610.61	− 2.96	0.35	8608.00	8605.03	2.97
8	9672.64	− 3.33	0.38	9669.69	9666.80	2.89
9	10 693.98	− 3.69	0.41	10 690.70	10 687.96	2.74
10	11 674.65	− 4.03	0.43	11 671.05	11 668.52	2.53
11	12 614.38	− 4.35	0.45	12 610.48	12 608.18	2.30
12	13 512.54	− 4.65	0.46	13 508.35	13 506.30	2.05
13	14 368.08	− 4.93	0.45	14 363.60	14 361.75	1.85
14	15 179.33	− 5.19	0.44	15 174.58	15 172.87	1.71
15	15 943.88	− 5.44	0.42	15 938.86	15 937.20	1.66
16	16 658.32	− 5.70	0.39	16 653.01	16 651.33	1.68
17	17 317.97	− 6.01	0.34	17 312.30	17 310.57	1.73
18	17 916.47	− 6.46	0.27	17 910.28	17 908.52	1.76
19	18 445.30	− 7.12	0.17	18 438.35	18 436.49	1.86
20	18 893.24	− 8.04	0.05	18 885.25	18 882.75	2.50
D_0	19 606.7	− 12.56	− 0.53	19 593.8	19 588.8[f]	5.0
R_e	3.01390	0.00076	− 0.00015	3.01451	3.01472[g]	− 0.00021
					3.01452[g]	− 0.00001
					3.01473[g]	− 0.00022

[a] *Ab initio* Born–Oppenheimer calculations obtained by the MR-CISD + Q_P level using 6Z/(56) basis set. See text for more detail.
[b] DBOC contribution calculated by the MR-CISD level using 8-3-1 reference space and cc-pwCVTZ basis.
[c] Relativistic correction obtained with the ic-MR-CISD level using cc-pwCV5Z basis set.
[d] Adiabatic calculations represent the sum of the BO, DBOC, and MVD contributions.
[e] Empirical stands for levels deduced by Coxon and Dickinson (Ref. 19) from the experimental spectroscopic data using a simultaneous fit of 9089 transitions involving X and A electronic states.
[f] Extrapolated from measurements of the convergence limit in the $^1\Sigma^+(X) \rightarrow {}^1\Pi(B)$ absorption bands by Vidal and Stwalley (Ref. 59).
[g] Empirical PEC equilibrium for ^{7}LiH accounting for DBOC corrections obtained from data by Uehara and Ogilvie (Ref. 21), Coxon and Molski (Ref. 60), and Molski (Ref. 61), respectively. See Sec. V for more details.

of the DBOC and ≈50 cm^{-1} in the case of the basis set dependence (Table III and Fig. 3 QZ/QZ results). This is because the change of the MVD contribution along the studied range of internuclear distances is much smaller than in the other two cases. Inspecting Fig. 7 one can see the nonmonotonic convergence of the MVD relativistic correction with respect to basis set. Similar behavior was already reported for correlation consistent bases[58] and indicates that the cc-pwCVXZ family of basis sets in their default contracted form are not well suited to converge the MVD relativistic energy contribution. However, it can be observed that the three curves corresponding to the different basis sets have quite similar shape shifted by ≈10^{-6} hartree, two orders lower than the magnitude of correction itself. Having in mind that the shift of the potential energy curve does not change the vibrational levels, it can be concluded that the icMR-CISD/cc-pwCV5Z method calculation is sufficient to provide the required accuracy.

The effects of the relativistic contribution on the vibrational levels are summarized in Table IV and also shown in Fig. 4. It is seen that the contribution is always smaller than 0.5 cm^{-1}, i.e., much smaller than the effect of either basis set or DBOC. Its inclusion is justified, but, due to its small size, the accuracy of the present approximation seems to be sufficient.

III. CALCULATION OF NONADIABATIC COUPLINGS AND THE β(R) FUNCTION

To evaluate the BO vibrational breakdown function β(R) defined by Eq. (3), the nonadiabatic coupling (NADC) vector $\langle\psi_0|\frac{\partial}{\partial R}|\psi_n\rangle$ needs to be obtained. Generally applicable analytic formula are available at the MCSCF and MR-CISD levels,[62,63] at the equation-of-motion coupled cluster level[64] and the time-dependent density functional theory level.[65] To keep up with the accuracy of the other contributions discussed above, in this work a MR-CISD wave function appears to be the appropriate choice. Efficient implementation of the NADC is available in the COLUMBUS program package,[38] which is based on analytical gradient techniques.[66] The advantage of the analytical approach is low computational cost: for a MR-CISD wave function, there is only a very little overhead compared to energy calculations.[62]

The calculations have been performed for the lowest seven excited states, three of $^1\Sigma^+$ symmetry and four of $^1\Pi$ symmetry. Only the $^1\Sigma^+$ states give nonzero contribution to the β(R) function, while the $^1\Pi$ states will be important for nonadiabatic effects on the rotational levels. This will be studied in a subsequent publication.[67] The cc-pwCV5Z basis set and the (8-3-1) CAS reference was adopted in all the MR-CISD calculations.

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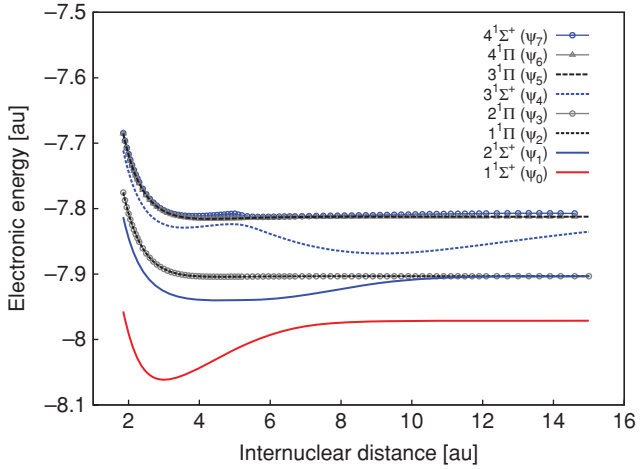


FIG. 8. MR-CISD adiabatic potential energy curves of the eight lowest electronic states of LiH calculated using the cc-pwCV5Z basis set and a 4 electrons in 15 orbitals (8-3-1) CAS reference wave function.

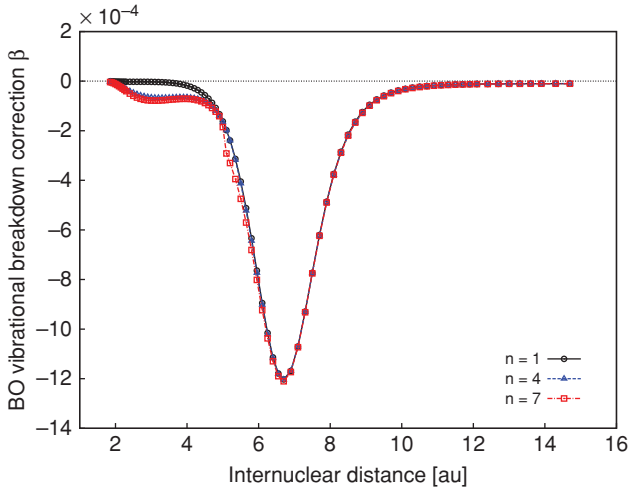


FIG. 10. Dependence of the BO breakdown correction β(R) of ^{7}LiH on the number of nonadiabatic coupling terms n used in the summation of Eq. (3).

Figure 8 depicts the potential energy surfaces for the seven excited states along with the ground state. As is well known, $2^1\Sigma^+$ is the first excited state, lying quite close to the ground state at larger distances. All other $^1\Sigma^+$ states are well separated energetically, therefore, it can be assumed that the $2^1\Sigma^+$ state will be the most important concerning the nonadiabatic effects, at least in the present approximation.

Figure 9 shows the NADC values for the first three $^1\Sigma^+$ states. In absolute terms $2^1\Sigma^+$ gives the largest values, but the two other states have also sizable contributions at some regions. Comparing Figs. 8 and 9, one can conclude that the maxima of absolute values of the NADC curves coincide with certain avoided crossings: $\langle\psi_0|\frac{\partial}{\partial R}|\psi_1\rangle$ has its maximum where the two states have an avoided crossing, while absolute values of $\langle\psi_0|\frac{\partial}{\partial R}|\psi_4\rangle$ and $\langle\psi_0|\frac{\partial}{\partial R}|\psi_7\rangle$ have extrema where ψ_4 and ψ_7 have avoided crossings. The NADC can also be sizable in the repulsive region where the states become nearly degenerate.

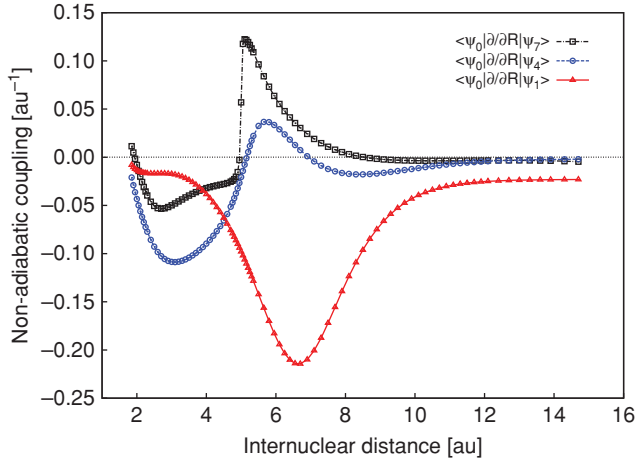


FIG. 9. MR-CISD nonadiabatic couplings of the ground $^1\Sigma^+$ (ψ_0) state with three excited $^1\Sigma^+$ (ψ_1, ψ_4, ψ_7) electronic states of ^{7}LiH calculated using the cc-pwCV5Z basis set and a 4 electrons in 16 orbitals (8-3-1) CAS reference wave function.

To evaluate β(R) according to Eq. (3), the NADC terms are divided by the energy difference of the states involved. This means that higher lying states will have decreasing importance for this quantity. Figure 10 demonstrates this effect: plotted are the cumulative curves of β(R) for the ^{7}LiH isotopologue with increasing number of states included. It is seen that the maximum of the absolute value for the β(R) function is associated with the first exited state (ψ_1), while the other states have negligible effect. On the other hand, at short distances the contributions from ψ_4 are of importance, although the corresponding values are an order of magnitude smaller than the other maxima at larger distances. A very small effect of ψ_7 can be observed on the curve at about 5 bohr, which hardly influences the vibrational levels. Further states are not expected to contribute to this β(R) function due to the large denominator in Eq. (3).

Finally, the mass-dependent β(R) functions[53] for all four isotopologues are plotted in Fig. 11. The curves clearly show that, due to its larger mass, the β(R) contribution will be smaller for the deuterated isotopologues, while contributions in the cases of ^{7}Li or ^{6}Li isotopologues are rather similar. In any event, it does not seem to be negligible for any of the four isotopologues of LiH. As will be seen in Sec. V, this contribution brings about a substantial improvement over the adiabatic treatment, modifying the vibrational levels by 1–2 cm^{-1}.

While preparing this manuscript, we became aware of a paper by Sauer *et al.*,[68] where the same vibrational nonadiabatic contributions have been evaluated for LiH along with the corresponding rotational quantity. Note that the same formula was used in that paper to evaluate the β(R) function at the MCSCF level using a large reference space and a basis set including up to f functions. In their Fig. 2, the same type of curve has been obtained as the one seen in Fig. 10 of this paper. The present calculations should be preferred over those of Sauer *et al.* due to the following factors: (a) a more accurate wave function in terms of electron correlation and basis set size has been used; (b) the β(R) function was obtained on a much denser grid; and (c) the corresponding vibrational

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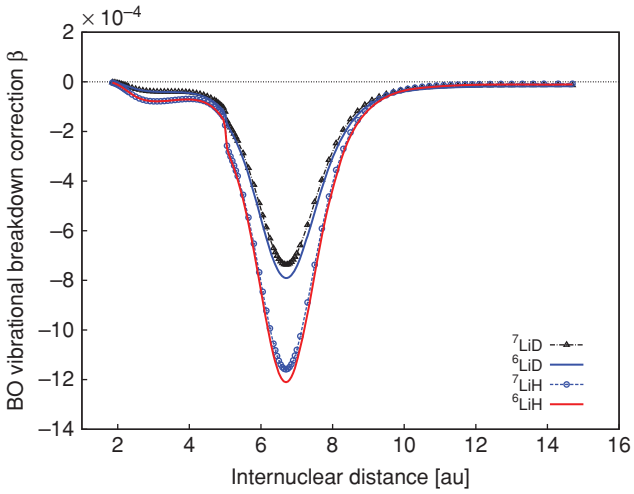


FIG. 11. Mass dependence of BO breakdown correction β(R) for studied isotopologues ^{7}LiH, ^{6}LiH, ^{7}LiD, and ^{6}LiD.

levels (see below) were calculated. Calculation of the rotational contribution is also in progress.[67]

IV. CALCULATION OF VIBRATIONAL ENERGY LEVELS

In order to compare with known experimental levels up to the dissociation limit,[19] a method for solving the nuclear equations had to be used that is accurate and reliable for all of these vibrational states.

Various methods can be used for a solution of the radial nuclear Schrödinger equation with the Hamiltonian equation (1). One is an algebraic perturbational approach originating from the Dunham work,[69] which has been further developed in the form of contact transformations[22,70] or the hypervirial perturbation theory.[71] Other methods which are considered as "numerically exact" ones rely on Numerov–Cooley wavefunction propagation methods[72–74] implemented in the computational code LEVEL by LeRoy[75] or on variational calculations.[76,77] A comparison of these approaches allows solving some convergence issues as has been discussed in Refs. 78 and 79. The Numerov–Cooley-type wavefunction propagation methods are known to provide very accurate results for vibrational levels up to the dissociation limit, but in its conventional form cannot be directly applied to the Hamiltonian of Eq. (1) because of the presence of the nonadiabatic correction function β(R) in the radial kinetic energy term. However, this term can be iteratively removed by a transformation discussed in Ref. 79 (see also references therein).

In this work nonperturbational variational calculations of the vibrational levels was employed following the technique described by Rey *et al.*[76,77] Variational-type methods do not require restrictions on the form of the kinetic energy terms. The Ritz functional is minimized using a set of trial primitive functions $\varphi_n(R; p_1, p_2, ...)$ by an optimization of linear or nonlinear parameters p_i. These primitive functions are usually chosen as eigenfunctions of some simple model. The maximum number of primitive functions is then increased until the convergence of computed energies is achieved.

As a first step *ab initio* electronic energies computed in Sec. II on a grid of 243 internuclear distances (R) in the range 1.85–15.0 bohr were fitted by an expansion involving an appropriate nuclear distance variable (z). Two variants of the analytical approximation of the *ab initio* radial functions involved in the Hamiltonian of Eq. (1) were tested. In the first one an expansion in the Dunham variable was used:

$$z_D = (R - R_e)/R_e, \tag{6}$$

using 30 expansion parameters[53] with an average rms deviation in the fit of ≈0.04 cm^{-1} slightly depending on the level of *ab initio* methods. The vibrational Hamiltonian of Eq. (1) was then expressed as a polynomial in creation and annihilation operators following Ref. 76. With this form of the Hamiltonian expansions it is quite natural to choose the harmonic oscillator wavefunctions as the primitive functions in the variational calculations. However, such wavefunctions yield poor convergence of vibrational levels lying near dissociation. In order to improve the convergence of the variational calculations, extended harmonic oscillator functions, φ_n (q;λ, κ) = ⟨q|n,λ,κ⟩, were used as initial trial wavefunctions. Here q is the dimensionless normal coordinate (q = $R_e z_D \sqrt{\mu\omega_e/\hbar}$) and |n,λ,$\kappa$⟩ are displaced squeezed Fock states (DSFS). These states are deduced from the usual Fock states |n⟩ by a unitary transformation given, for example, in Refs. 80–82. Dimensionless nonlinear parameters λ and κ control and optimize the convergence of the energy values. Note that by using the basis set of DSFS, the matrix elements of the Hamiltonian can be computed analytically[77] without loss of precision. For the current practical applications to the LiH molecule, the parameters λ and κ were set to 5.5 and 0.65, respectively. The maximum number of primitive functions used was 75 and this was applied for all four isotopomers. The error of the highest vibrational energy level due to the vibrational basis set truncation was estimated to be less than 5 × 10^{-4} cm^{-1}(estimated from calculations using up to 90 primitive functions).

In order to control the error related to the analytical representation of *ab initio* radial functions, the potential function was also expanded in a nonlinear Ogilvie–Tipping variable:

$$z_{\mathrm{OT}} = 2(R - R_e)/(R + R_e). \tag{7}$$

Twenty-two expansion parameters were included with a similar rms deviation (compared to the Dunham variable expansion) in the fit of ≈0.04 cm^{-1}. In order to compute matrix elements of the Hamiltonian, eigenfunctions of the Kratzer oscillator were used as the primitive functions. The maximum discrepancy in computed vibrational levels using these two independent approximations of *ab initio* potentials was 10^{-2} cm^{-1} up to $\mathbf{v} = 20$. Finally, the convergence of the variational calculations was checked via comparison with a Numerov–Cooley solution of the radial equation for simple cases where the β(R) function was not included and the LEVEL code could be easily applied. The difference with the variational calculations was only 10^{-5} cm^{-1} for all vibrational states, which confirmed the convergence estimates.

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TABLE V. Comparison of *ab initio* and empirical vibrational levels (in cm^{-1}) for different isotopologues of LiH. Nuclear masses and the BO breakdown correction $\beta(R)$ have been used in the vibrational levels calculations.

v	^{7}LiH[a]	Calc.–Emp.[b]	^{6}LiH[a]	Calc.–Emp.[b]	^{7}LiD[a]	Calc.–Emp.[b]	^{6}LiD[a]	Calc.–Emp.[b]
1	1360.58	0.88	1374.24	0.88	1029.62	0.56	1048.07	0.56
2	2676.11	1.55	2702.53	1.58	2033.67	1.01	2069.62	1.01
3	3947.53	2.06	3985.81	2.11	3012.57	1.37	3065.10	1.36
4	5175.69	2.41	5224.96	2.49	3966.69	1.64	4034.91	1.62
5	6361.35	2.63	6420.76	2.72	4896.40	1.83	4979.44	1.80
6	7505.19	2.71	7573.90	2.82	5802.03	1.95	5899.02	1.89
7	8607.73	2.70	8684.91	2.82	6683.88	1.99	6793.97	1.91
8	9669.38	2.58	9754.19	2.72	7542.23	1.98	7664.59	1.88
9	10 690.35	2.39	10 781.96	2.54	8377.30	1.91	8511.09	1.78
10	11 670.64	2.13	11 768.17	2.29	9189.27	1.78	9333.67	1.64
11	12 610.00	1.82	12 712.54	2.00	9978.28	1.62	10 132.46	1.45
12	13 507.79	1.50	13 614.39	1.70	10 744.40	1.41	10 907.52	1.22
13	14 362.96	1.20	14 472.53	1.44	11 487.64	1.17	11 658.83	0.96
14	15 173.83	0.97	15 285.16	1.24	12 207.92	0.91	12 386.29	0.67
15	15 938.00	0.81	16 049.64	1.11	12 905.04	0.63	13 089.68	0.38
16	16 652.03	0.71	16 762.26	1.03	13 578.73	0.36	13 768.63	0.10
17	17 311.21	0.65	17 417.89	0.99	14 228.51	0.10	14 422.62	− 0.16
18	17 909.13	0.61	18 009.56	0.98	14 853.77	− 0.12	15 050.92	− 0.38
19	18 437.20	0.72	18 527.94	1.19	15 453.65	− 0.31	15 652.54	− 0.56
20	18 884.19	1.45	18 960.88	2.19	16 027.04	− 0.44	16 226.20	− 0.68
21	(19 236.07)[c]		19 293.34	4.35	16 572.50	− 0.53	16 770.23	− 0.75
22	(19 476.21)[c]		(19 507.64)[c]		17 088.20	− 0.58	17 282.50	− 0.78
23					17 571.81	− 0.60	17 760.33	− 0.78
24					18 020.43	− 0.57	18 200.36	− 0.71
25					18 430.44	− 0.45	18 598.37	− 0.48
26					18 797.39	− 0.06	18 949.24	0.16
27					19 115.93	0.91	(19 246.86)[c]	
28					(19 379.84)[c]		(19 484.22)[c]	
29					(19 582.07)[c]		(19 653.44)[c]	
30					(19 715.25)[c]			
ZPE	698.45		705.65		525.07		534.68	

[a]The *ab initio* potential function includes BO, DBOC, and MVD contributions.
[b]Empirically determined vibrational levels are taken from Ref. 19.
[c]Numbers in parenthesis correspond to extrapolated *ab initio* levels up to dissociation.

V. DISCUSSION AND CONCLUSIONS

The experimentally known vibrational levels of the electronic ground state of LiH span an energy range covering about 98.6% of the dissociation energy. The energies of high-lying vibrational levels very close to the dissociation limit with vibrational quantum numbers $\mathbf{v} = 21$, 20, 7, and 27 for ^{6}LiH, ^{7}LiH (the most abundant isotopologue), ^{6}LiD, and ^{7}LiD species, respectively, have been deduced from spectroscopic experiments.[19] Most of the data for the highly excited vibrational levels were assigned, however, from transitions in the $A^1\Sigma^+ - X^1\Sigma^+$ band systems, while the infra-red bands corresponding to transitions observed within the X electronic ground state were obtained up to $\mathbf{v} = 3$, 6, 3, and 6 for the four isotopomers mentioned above, respectively.[19,60] These data will be used, as well as their empirical extrapolations, to compare with the present theoretical predictions.

Based on the detailed analysis above, the best *ab initio* potential energy curve is composed of: (i) an *ab initio* nonrelativistic potential obtained at the MR-CISD+Q_P/6Z/(56) level, (ii) MVD relativistic corrections obtained at the icMR-CISD/cc-pwCV5Z level, and (iii) a DBOC obtained at the MR-CISD/cc-pwCVTZ level. Note that the first two contributions give the best calculated BO potential, while the third one introduces the adiabatic correction to it.

First, the properties of this potential energy function are discussed, starting with the corresponding equilibrium geometry and the dissociation energy. Though R_e and D_e are not really observable quantities it is instructive to compare our *ab initio* values with available empirically determined counterparts obtained through model dependent fits to the experimental data.[19,21,22,59–61]

The nonrelativistic *ab initio* potential gives an equilibrium bond length R_e(BO, nonrelativistic) = 3.01390 bohr. Inclusion of the relativistic correction changes this only slightly, giving a final theoretical BO value of R_e(BO, *ab initio*) = 3.01375 bohr. This is in excellent agreement with the empirically determined value of R_e(BO, empirical) = 3.01394 bohr (using 1 bohr = 0.529177248 Å). This latter value has been reported in several papers (Watson,[22] Uehara and Ogilvie,[21] Molski,[61] and Coxon and Molski[60] all agreeing to six digits given above) and has been obtained by the simultaneous fit of rovibrational transitions of the four isotopologues. The deviation of the *ab initio* and empirical value is only

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TABLE VI. Comparison of *ab initio* and empirical vibrational levels (in cm^{-1}) for different isotopologues of LiH. Atomic masses have been used in the vibrational level calculations.

v	^{7}LiH[a]	Calc.–Emp.[b]	^{6}LiH[a]	Calc.–Emp.[b]	^{7}LiD[a]	Calc.–Emp.[b]	^{6}LiD[a]	Calc.–Emp.[b]
1	1360.30	0.59	1373.96	0.60	1029.51	0.45	1047.95	0.45
2	2675.56	1.00	2701.98	1.03	2033.46	0.79	2069.39	0.78
3	3946.73	1.26	3985.01	1.30	3012.25	1.05	3064.76	1.03
4	5174.66	1.38	5223.92	1.45	3966.27	1.22	4034.47	1.18
5	6360.11	1.38	6419.51	1.47	4895.88	1.32	4978.90	1.26
6	7503.75	1.28	7572.45	1.38	5801.43	1.35	5898.39	1.26
7	8606.13	1.09	8683.29	1.21	6683.20	1.31	6793.26	1.21
8	9667.63	0.83	9752.43	0.97	7541.48	1.23	7663.80	1.09
9	10 688.48	0.51	10 780.07	0.67	8376.48	1.09	8510.24	0.93
10	11 668.68	0.16	11 766.21	0.33	9188.39	0.91	9332.76	0.72
11	12 607.98	− 0.21	12 710.53	− 0.01	9977.35	0.69	10131.50	0.48
12	13 505.75	− 0.55	13 612.36	− 0.32	10 743.43	0.44	10906.51	0.21
13	14 360.93	− 0.82	14 470.53	− 0.56	11 486.63	0.16	11657.79	− 0.08
14	15 171.87	− 1.00	15 283.23	− 0.69	12 206.88	− 0.13	12385.23	− 0.39
15	15 936.14	− 1.06	16 047.83	− 0.70	12 904.00	− 0.42	13088.61	− 0.69
16	16 650.32	− 1.01	16 760.61	− 0.61	13 577.68	− 0.69	13767.56	− 0.97
17	17 309.68	− 0.89	17 416.44	− 0.45	14 227.48	− 0.93	14421.57	− 1.21
18	17 907.81	− 0.71	18 008.33	− 0.24	14 852.76	− 1.13	15049.91	− 1.40
19	18 436.12	− 0.37	18 526.95	0.20	15 452.68	− 1.28	15651.58	− 1.53
20	18 883.35	0.60	18 960.14	1.44	16 026.13	− 1.36	16225.30	− 1.58
21	(19 235.47)[c]		19 292.85	3.86	16 571.65	− 1.38	16769.40	− 1.58
22	(19 475.88)[c]		(19 507.41)[c]		17 087.42	− 1.36	17281.76	− 1.52
23					17571.12	− 1.29	17 759.69	− 1.42
24					18019.83	− 1.17	18 199.81	− 1.26
25					18429.93	− 0.96	18 597.92	− 0.93
26					18796.98	− 0.48	18 948.88	− 0.20
27					19115.60	0.58	(19 246.58)[c]	
28					(19379.59)[c]		(19 484.03)[c]	
29					(19581.91)[c]		(19 653.33)[c]	
30					(19714.95)[c]			
ZPE	698.30		705.50		525.01		534.62	

[a]The *ab initio* potential function includes BO, DBOC, and MVD contributions.
[b]Empirically determined vibrational levels are taken from Ref. 19.
[c]Numbers in parenthesis correspond to extrapolated *ab initio* levels up to dissociation.

0.0002 bohr, which is probably at the limit of theoretical predictions nowadays. However, such level of accuracy is known to be necessary for an accurate calculation of rotational levels for light molecules.[83] Note that Coxon and Dickinson[19] reported a somewhat different value, $R_e = 3.01524$ bohr, a clear outlier among empirical determinations. A possible explanation for this discrepancy is that while others included only the electronic ground state in the empirical determination of the equilibrium geometry, Coxon and Dickinson[19] have used also transitions between the ground state and the first excited state and fitted the corresponding surfaces simultaneously. Since the potential curve of the A state is very flat with a poorly defined R_e, this might have influenced also the results for the ground state.

Considering also the DBOC, the equilibrium bond length becomes mass dependent and increases; for ^{7}LiD $R_e(^7\text{LiD})$ is 3.01417 bohr, while for the light isotopologue ^{7}LiH it is even longer [$R_e(^7\text{LiH}) = 3.01451$ bohr]. Empirically obtained values for ^{7}LiH can be found in Table IV, while those for ^{7}LiD are 3.01438, 3.01429, and 3.01439 bohr obtained from the data by Uehara and Ogilvie,[21] Coxon and Molski,[60] and Molski,[61] respectively. Not just the directions but also the magnitude of these changes are in qualitative agreement with the listed empirical findings.

A good quality of our *ab initio* R_e value is confirmed by rotational predictions, the details of which will be reported in the forthcoming paper.[67] The rms deviation for rotational spacings dE = E(v, J = 1) −E(v, J = 0) between our *ab initio* calculations and empirical results by Coxon and Dickinson[19] is 0.010 cm^{-1} (with nuclear masses) and 0.006 cm^{-1} (with atomic masses) over all available vibrational ^{7}LiH states from v = 0 to v = 20. Note that for high vibrational states with v > 6 this falls within the experimental uncertainty of experimental measurements.

The *ab initio* dissociation limit including DBOC is $D_e(^7\text{LiH}) = 20292$ cm^{-1} and $D_e(^7\text{LiD}) = 20300$ cm^{-1}. These compare very well with empirical extrapolations by Vidal and Stwalley[59] [$D_e(^7\text{LiH}) = 20287$ cm^{-1} and $D_e(^7\text{LiD}) = 20293$ cm^{-1}]. In both cases D_e of LiD is higher than that of LiH, by 5 cm^{-1} in the present calculations and by 6 cm^{-1} in Vidal's empirical determination.

The *ab initio* data presented in this work allow the calculation of vibrational levels including nonadiabatic contributions. Two different *ansatze* have been used for this

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TABLE VII. Summary of rms deviations between *ab initio* and empirical values of vibrational energies.

		^{7}LiH	^{6}LiH	^{7}LiD	^{6}LiD
$\mathbf{v}_{max}$[a]		20	21	27	26
E^{v}_{max}[b]		19 582	19 998	19 641	19 484
D_e		20 292	20 292	20 300	20 300
E^{v}_{max}/D_e (%)		96.5	98.6	96.8	96.0
Ab initio potential[c]	Masses		rms[d] (cm^{-1})		
BO	Nuclear	6.26	6.95	2.80	2.76
BO + DBOC	Nuclear	2.24	2.66	1.29	1.19
BO + DBOC + β(R)	Nuclear	1.79	2.14	1.18	1.15
BO + DBOC	Atomic	0.91	1.22	1.01	1.07

[a]Maximum value of vibrational quantum number for empirically extrapolated energies as given by Coxon and Dickinson (Ref. 19).
[b]Corresponding maximal vibrational energies (including ZPE) involved in the rms comparisons.
[c]Components of the *ab initio* potential, see text for precise explanation.
[d](Calc.–Emp.) using *ab initio* values of Tables IV and V.

purpose: (a) either the kinetic nonadiabatic β(R) function obtained at the MR-CISD/cc-pwCVTZ level was included in the Hamiltonian of Eq. (1) along with nuclear masses or (b) atomic masses were used in the Hamiltonian but the nonadiabatic function was not considered. As has been discussed by Kutzelnigg,[84] this latter choice also includes nonadiabatic effects in an approximate way.

All vibrational levels for all four isotopologues obtained with the two strategies are listed in Tables V–VI. Table VII presents a summary of the results along with the rms deviations between the *ab initio* predictions and all available empirically determined values by Coxon and Dickinson.[19] This comparison covers about 96–98% of the dissociation energy. The results are very pleasing; the calculated BO potential (without mass-dependent DBOC corrections) results in vibrational levels that have rms deviations of about 6–7 cm^{-1} for LiH and ~3 cm^{-1} for LiD. If also the DBOC is accounted for, the rms deviation falls down to ~2 cm^{-1} for LiH and ~1 cm^{-1} for LiD. Finally, by accounting for the nonadiabatic effects by *ansatz* (a) [inclusion of the β(R) function] the rms deviation is smaller than 2 cm^{-1}. For *ansatz* (b) it is further reduced, being only slightly over 1 cm^{-1}.

The final theoretical PEC[53] represents a drastic improvement over previous predictions of vibrational levels for all four isotopologues, with an average error of only 1–2 cm^{-1}. This is to be compared with the best previous *ab initio* PEC by Li and Paldus,[17] which resulted in vibrational levels differing by as much as 30 cm^{-1} for highly excited vibrational states. The most crucial methodological improvements introduced in this work include: (i) the use of a (56) CBS extrapolation for the *ab initio* energies and (ii) the inclusion of DBOC corrections, both resulting in up to a 10 cm^{-1} improvement for higher vibrational levels (for ^{7}LiH at $\mathbf{v} = 20$). The remaining error of 1–3 cm^{-1} can most probably be assigned to NAD effects, i.e., coupling with excited electronic states. The two schemes used here are not capable of recovering this effect completely—the use of atomic masses is "just" a pragmatical, approximate method (see Ref. 84), while the β(R) function represents only one of the terms from the complete treatment of nonadiabatic effects, and the second derivative terms in the Bunker–Moss–Schwenke formulation of NAD[5,25,34] corresponding to the $U^{(2)}(R)$ contribution in Eq. (1) is not accounted for. There is not enough information in the literature to reveal the magnitude of this term for LiH. Other sources of errors could be residual inaccuracy of the BO term, continuous spectrum contribution to β(R), and insufficient description of relativistic effects at larger internuclear distances. Therefore, it remains an open question whether the remaining 1–2 cm^{-1} average deviation is caused by further nonadiabatic effects or remaining inaccuracies of other components of the present treatment. Calculation of the second derivative term is rather involved, but this work is currently being undertaken. Also, computation of rotational–vibrational transitions for the four isotopologues and their extended comparison with the available spectroscopic data will be the subject of a forthcoming paper.[67]

To summarize, the PEC presented in this work represents to our knowledge the best currently available *ab initio* potential function for more than a two-electron system that allows a uniformly accurate prediction of vibrational energy levels up to the dissociation limit. It must be emphasized, however, that application of this or similar procedure to larger molecules heavily depends on future advances in computer technology, including both software and hardware components.

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